ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · B2 · DAY, NIGHT AND MOON

Earth, Sun and Moon

★☆☆
Grade 3

Learning goals

- Recall key facts.
- Name Moon phases.

1 Fill in the gaps. [4]

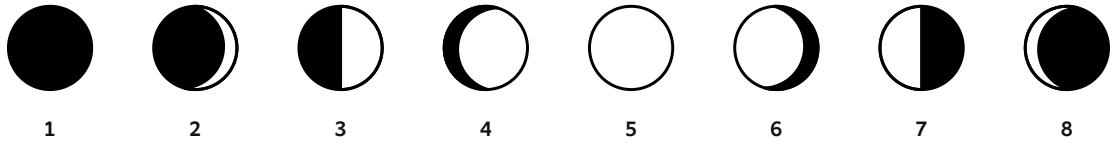
WORD BANK

Sun 24 year week Earth night

The Earth spins once every 24 hours. This makes day and night.

The Earth goes around the Sun once a year. The Moon goes around the Earth.

2 Name the Moon phases. [4]



about 29½ days from one new moon to the next

1	2	3	4	5	6	7	8
new moon	waxing crescent	first quarter	waxing gibbous	full moon	waning gibbous	last quarter	waning crescent

3 True or false? [3]

Statement	True	False	If false, correct it
1. The Sun moves around the Earth every day.		✔	The Earth spins; the Sun only seems to move.
2. The Moon makes its own light.		✔	It reflects sunlight.
3. Shadows are shortest at midday.	✔		

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · B2 · DAY, NIGHT AND MOON

Explaining patterns

★★☆
Grade 3

Learning goals

- Explain day and night and seasons.

1 **When it is day in Bangladesh, it is night in America. Why?** [2]

Bangladesh faces the Sun while America is on the other side of the spinning Earth, facing away.

2 **Why is it hotter in summer than in winter?** [2]

In summer the Sun is higher in the sky, so its light is stronger.

3 **Draw a shadow stick at 9 am, 12 noon and 4 pm.** [3]

Shadow stick

Long shadow in the morning (pointing west), short at noon, long in the afternoon (pointing east).

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

INVESTIGATE · B2 · DAY, NIGHT AND MOON

Make a shadow stick

★★☆
Grade 3

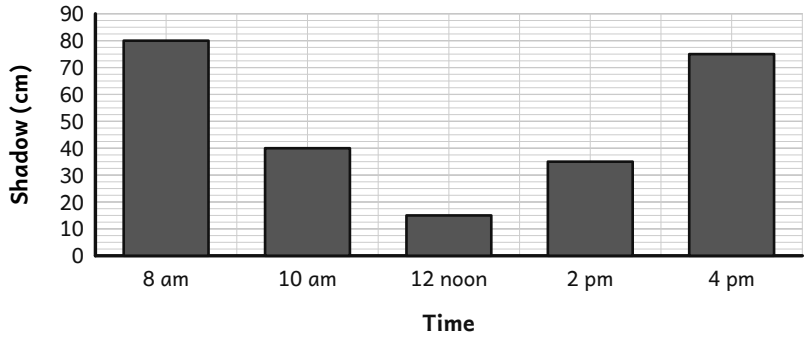
Learning goals

- Measure shadows through the day.
- Find a pattern.

Push a stick into the ground in a sunny place. Measure the length of its shadow every two hours.

1 **Here are Class 3A's results. Draw a bar chart.** [3]

Time	8 am	10 am	12 noon	2 pm	4 pm
Shadow (cm)	80	40	15	35	75



2 **When was the shadow shortest? Why?** [2]

At 12 noon, because the Sun was highest in the sky.

3 **Predict the shadow at 5 pm.** [1]

longer than 75 cm

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · B2 · DAY, NIGHT AND MOON

The Moon and the fishers

★★★
Grade 3

Learning goals

- Learn why the Moon matters to people.

Science | Work | Culture

Fishers on the coast of Bangladesh watch the Moon carefully. At **full moon** and **new moon**, the tides are very high and very low. These are called **spring tides**. The fishers plan when to go out and when to bring their boats in.

Families also watch for the thin **crescent moon** after the new moon. When it is seen, a new month of the Islamic calendar begins, and everyone knows when **Eid** will be celebrated.

Based on tides and moon sighting in Bangladesh.

1 **Why do fishers watch the Moon?** [2]

The Moon causes the tides; they plan their fishing trips with the tides.

2 **Which Moon phase begins a new month in the Islamic calendar?** [1]

the new crescent moon

3 **Draw the Moon tonight, or the last time you saw it. Which phase was it?** [2]

The Moon I saw

Any phase, labelled.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REVIEW AND CHALLENGE · B2 · DAY, NIGHT AND MOON

Show what you know

★★★
Grade 3

Learning goals

- Use Earth and Moon words.

1 Find the words.

J	S	F	E	G	R	V	G	U
T	N	P	N	W	A	D	A	O
K	C	I	I	W	S	A	F	D
F	Y	K	G	N	I	Y	L	W
G	R	D	I	H	D	M	R	M
T	I	D	E	S	T	K	U	O
E	C	S	E	A	S	O	N	O
O	R	B	I	T	L	U	D	N
U	K	R	T	H	F	C	K	O

day moon night orbit season spin tide

2 Why do we have day and night? [1]

○ the Sun goes around the Earth
● **the Earth spins**
○ clouds cover the Sun

3 How long does the Earth take to go around the Sun? [1]

○ one day
○ one month
● **one year**

4 Challenge. Use a torch and a ball to explain day and night to a friend. Draw how you would set it up. [2]

My model

Torch shining on a ball; turn the ball slowly; the lit side is day.